nelsonnet

Name: ______________________________

WORKSHEET

Complex numbers

Do not be fooled by those who see you as just another consumer in a market of undifferentiated possibilities, where choice itself becomes the good, …

13	4	1	19	2	17	12

18	15	18	10	9	15

16	19	11	18	17	12

,

11	13	14

15	18	16	6	19	3	17	8	1	19

19	5	9	19	10	8	19	13	3	19

14	8	15	9	2	11	3	19	15

17	10	18	17	7

Pope Benedict XVI

Questions

Simplify the following:

1 $5i + 2i + 4i$

2 $19i - 13i$

3 i^3

4 i^{-8}

5 $5i\,(3i - 7)$

6 $\sqrt{-49}$

7 $\frac{12i - 8i}{5i}$

8 $\frac{27x^3i^5 - 63x^2i^7}{(\sqrt{3}i)^2}$

9 $6i^3 \times 2i^2 - 7i \times 4i^{-4}$

10 $(2i + 9)\,(3 - 4i)$

11 $(5 + 7i)^2$

Solve the following:

12 $x^2 + 7 = 0$

13 $2x^2 + 50 = 0$

14 $4x^3 + 36x = 0$

15 $5x^2 + 6x + 5 = 0$

16 $8x - 3x^2 = 9$

State the nature (real/complex or equal) of the roots of the following equations:

17 $2x^2 - 7x + 5 = 0$

18 $9x^2 - 12x = -4$

19 $-3x^2 + 5x - 4 = 0$

Solutions

A $-24 + 70i$

B $\frac{4 \pm i\sqrt{11}}{3}$

C $-i$

D $x = 0$ or $x = \pm 3i$

E Complex

H $\frac{4}{5}$

I $-3x^2(3x + 7)i$

J $7i$

L $6i$

N $\pm 5i$

O 1

P $-16i$

R $35 - 30i$

S $\frac{-3 \pm 4i}{5}$

T Real

U Equal

V $11i$

X $-15 - 35i$

Y $\pm i\sqrt{7}$

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